

AI Agent Battle — Tic-Tac-Toe

Assignment X_02 | AI/ML Laboratory

This report investigates two autonomous Tic-Tac-Toe agents using Minimax with Alpha-Beta pruning. NEXUS uses a tactical threat-oriented heuristic (H1), while TITAN uses a positional/flexible-line heuristic (H2). The main battle uses depth 3 for both agents and alternates the starting player.

1. Implementation

The system is split into a game engine, search engine, heuristic functions, agent definitions, and experiment runner. Terminal wins/losses/draws receive exact scores of +100/-100/0. At a non-terminal depth limit, the selected heuristic evaluates the board. Alpha-Beta pruning skips branches that cannot affect the current Minimax decision. Search statistics count evaluated and pruned nodes.

2. Depth Experiment

To isolate search depth more cleanly, NEXUS is the measured agent: its depth is changed from 1 to 4 while its Minimax + Alpha-Beta algorithm and H1 heuristic remain fixed. TITAN is kept as the fixed opponent at depth 3. The starting player continues to alternate. Thus, the principal experimental variable is NEXUS's search depth.

Depth	Games	NEXUS wins	TITAN wins	Draws	NEXUS avg nodes	NEXUS avg pruned	NEXUS avg ms
1	10	0	1	9	22.1	0.0	3.522
2	10	0	0	10	120.5	0.0	4.301
3	10	0	0	10	338.2	281.1	6.485
4	10	0	0	10	1199.7	610.3	15.193

The measured results show the computational trade-off directly: deeper search expands the number of positions considered and increases execution time, while also increasing the amount of pruning opportunity. Game outcomes are empirical and depend on the implemented heuristics and tie-breaking.

3. Ten-Game AI Battle

Game	First	Winner	Moves	N Nodes	T Nodes	N Pruned	T Pruned	Time ms
1	NEXUS	DRAW	9	402	258	379	206	7.083
2	TITAN	DRAW	9	273	393	191	380	6.960
3	NEXUS	DRAW	9	424	264	357	200	7.603
4	TITAN	DRAW	9	244	379	220	394	6.433
5	NEXUS	DRAW	9	402	258	379	206	7.210
6	TITAN	DRAW	9	270	404	194	369	6.849
7	NEXUS	DRAW	9	414	259	359	205	6.635
8	TITAN	DRAW	9	273	393	191	380	7.933
9	NEXUS	DRAW	9	436	230	345	234	6.745
10	TITAN	DRAW	9	273	393	191	380	6.930

4. Analysis

- 1. Did increasing search depth change the AI's decisions?** The depth results show that changing NEXUS's depth changes the search horizon and can change outcomes/decisions when shallow evaluation cannot see future tactical consequences.
- 2. Did deeper search increase execution time?** Yes in the measured runs: the average NEXUS search time rises as the search horizon expands.
- 3. Did the number of evaluated nodes increase?** The measured NEXUS node counts increase substantially with depth, reflecting the growing search tree.
- 4. Did Alpha-Beta reduce the number of nodes explored?** Yes. The implementation records pruned branches whenever alpha-beta bounds make remaining sibling branches irrelevant to the current Minimax choice.
- 5. Did the two agents make different decisions?** They can, because H1 and H2 assign different values to non-terminal positions. Equal-valued alternatives are randomized only among equally good moves.
- 6. How did their heuristics influence behaviour?** H1 emphasizes immediate threats; H2 emphasizes flexible winning-line potential plus center and corner control.
- 7. Did first-player advantage appear?** The starting player alternated. The results can therefore be compared by starting side rather than assuming that every win is caused by the heuristic.
- 8. Which agent won more games?** NEXUS = 0, TITAN = 0, DRAW = 10.
- 9. Were there many draws?** There were 10 draws out of 10 games.
- 10. Did the agent that won more require more computation?** Main-battle average nodes were NEXUS 341.1 and TITAN 323.1; average pruned counts were NEXUS 280.6 and TITAN 295.4; average total game time was 7.038 ms.

5. Conclusion

The experiment demonstrates the trade-off between deeper adversarial search and computational cost. Alpha-Beta pruning improves search efficiency without changing the fundamental Minimax objective, while heuristic design influences which non-terminal positions are preferred. The observed winner counts are evidence for this particular experimental configuration, not a universal claim that one heuristic is always superior.